

EVICHAIN

Complete Project Study Guide & Operator Manual

Architecture, Security, Workflows, APIs, Debugging, Beginner Manual, and Jury Defense

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EviChain Complete Project Study Guide

Architecture, Security, Workflows, APIs, Debugging, Beginner Operator Manual, and Jury Defense

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1. Executive Overview

What is EviChain?

EviChain is an enterprise-grade Digital Evidence Management System (DEMS) and Cryptographic Chain of Custody platform designed for law enforcement agencies, forensic examiners, prosecutors, and public auditors.

In traditional physical and digital policing, evidence handling suffers from severe vulnerabilities:

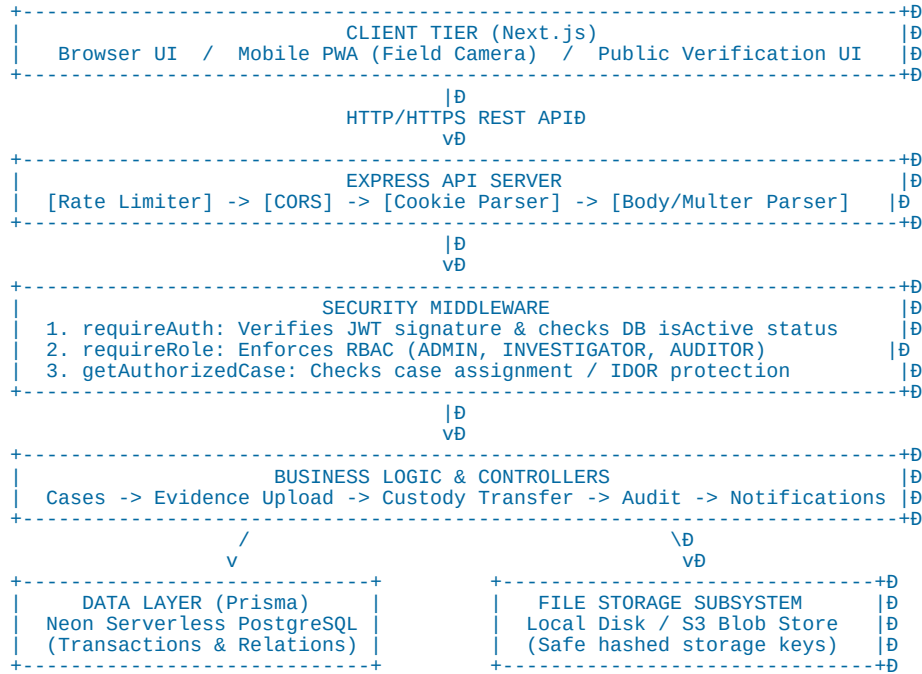
- Unmonitored physical custody transfers leading to lost or swapped USB drives/hard drives.
- Lack of cryptographic verification, allowing malicious actors to alter files unnoticed.
- Inconsistent paper-based custody logs that can be backdated, misplaced, or forged.
- Lack of public transparency, making it impossible for citizens or independent legal teams to verify whether an evidence record has been altered without compromising case confidentiality.

EviChain solves these fundamental challenges through a cryptographic, append-only, role-governed digital pipeline:


```
%      % % % integrity.service.ts      # Non-destructive relational consistency auditor
%      % % % notification.service.ts    # Notification dispatch & deduplication
%      % % % pdf.service.ts             # PDFKit court report generator
% % % tests/                            # 13 Automated Test Suites (194 tests)
```

3. Architecture & Request Lifecycle

High-Level Architecture Flow



Frontend vs Backend Responsibility

- Frontend (Next.js):** Responsible purely for presentation, client-side route guards, responsive mobile UX, offline camera capture, and optimistic caching. The frontend is **untrusted**.
- Backend (Express + Prisma):** Sole source of truth. Cryptographic hashing, authorization checks, role verification, database transactions, and audit trail recording are strictly executed server-side. Even if a compromised client sends a fabricated SHA-256 hash or claims ownership of evidence, the backend rejects it.

4. Database & Domain Model (13 Verified Entities)

EviChain utilizes a PostgreSQL schema defined in `server/prisma/schema.prisma`. Every entity has been verified:

#	Entity Model	Primary Key	Key Foreign Keys & Indexes	Purpose in EviChain
1	User	id (UUID)	Indexes: role, isActive	Core identity, authentication credentials (bcrypt hash), role, and active status
2	Session	id (UUID)	FK: userId -> User.id. Indexes: token, expiresAt	Server-tracked refresh sessions enabling remote device logout and revocation
3	Case	id (UUID)	FK: leadUserId -> User.id. Indexes: status, leadUserId	Legal case dossier grouping evidence, lead investigator assignment, priority
4	Evidence	id (UUID)	FK: caseId, collectedById, currentCustodianId. Indexes: sha256, storageKey	Evidence asset master record, file size, MIME type, SHA-256 hash, storage key
5	CustodyEvent	id (UUID)	FK: evidenceId, actorUserId, fromUserId, toUserId. Indexes: [evidenceId, timestamp]	Immutable chain of custody ledger events (CREATED, TRANSFERRED, ACCESSED)
6	AuditLog	id (UUID)	FK: actorUserId -> User.id. Indexes: [resourceType, resourceId]	System-wide append-only audit trail recording every state change and IP address
7	NotificationPreference	id (UUID)	FK: userId -> User.id (Unique, 1-to-1)	Per-user alert toggles for evidence uploads, transfers, security alerts
8	CaseComment	id (UUID)	FK: caseId, userId, parentId (Self-relation for replies)	Threaded collaboration discussions on case files with soft-delete support
9	CommentMention	id (UUID)	FK: commentId, userId	User @mentions extracted from comments to trigger targeted notifications
10	EvidenceAnnotation	id (UUID)	FK: evidenceId, userId. Indexes: [evidenceId, createdAt]	Non-destructive coordinate markers (points, arrows, regions) on evidence images
11	Notification	id (UUID)	FK: userId -> User.id. Unique: dedupeKey	In-app notification alerts with deduplication keys preventing alert spam
12	SearchPreset	id (UUID)	FK: userId -> User.id	User-saved search criteria and query presets
13	SystemSetting	id (UUID)	Unique: key	Global system configuration parameters (retention policies, storage mode)

5. Module-by-Module Technical Deep Dive

- **MODULE 1 — Repository & Architecture Scaffold**: Monorepo structure, unified typing, Next.js App Router, Express TypeScript server, CORS & 10MB JSON limits.
- **MODULE 2 — Database and Storage Foundation**: Persistent storage abstraction (Local Disk & S3), safe UUID storage keys preventing path traversal.
- **MODULE 3 — Case Management**: Multi-case dossier creation, lead assignment, lifecycle tracking, and IDOR protection.
- **MODULE 4 — Evidence Upload and SHA-256 Integrity**: Chunked streaming file intake with on-the-fly cryptographic fingerprinting and automatic compensation unlinking on failure.
- **MODULE 5 — Custody Transfer and Secure Access**: Legally binding chain of custody transfers with digital signatures and 5-minute access log deduplication.
- **MODULE 6 — Zero-Knowledge Public Verification**: Public and defense counsel verification of evidence integrity without exposing confidential case dossiers or officer names.
- **MODULE 7 — Reports, Audit Export & Compliance Intelligence**: Court-admissible Chain of Custody PDF generation and CSV export with spreadsheet formula injection sanitization.
- **MODULE 8 — Notifications and User Preferences**: Real-time event notifications for custody changes, mentions, and case assignments with deduplication keys.
- **MODULE 9 — Search, Command Palette and Discovery**: Instant keyboard-driven navigation (Cmd+K) and multi-attribute evidence search scoped to user authorization.
- **MODULE 10 — Admin and Profile Management**: User lifecycle management, role modifications, session revocation, and last-admin logout defense.
- **MODULE 11 — Mobile PWA & Offline Field Capture**: Crime scene evidence capture on mobile devices with rear environment camera and IndexedDB offline queue fallback.
- **MODULE 12 — Collaboration, Threaded Notes & Image Annotations**: Collaborative investigation via threaded case discussions and non-destructive visual coordinate annotations.
- **MODULE 13 — Final Integration, Security Hardening & Readiness**: Cross-module security sweep, mid-session deactivation enforcement, deep health probes, and 194 passing automated tests.

6. Complete API Reference Matrix

Resource	Group	Method	Path	Auth Required	Allowed Roles	Summary / Purpose
Health	GET	/health	No	Public	Lightweight liveness probe	
	GET	/health/deep	No	Public	Deep readiness check (PostgreSQL + Storage)	
Auth	POST	/auth/register	No	Public	User self-registration	
	POST	/auth/login	No	Public	Returns access token + HTTP-only refresh cookie	
	POST	/auth/refresh	No (Cookie)	Public	Rotates access token via refresh token	
	POST	/auth/logout	Yes	All	Clears session cookie and invalidates session	
	POST	/auth/logout-all	Yes	All	Revokes all active sessions for current user	
Cases	GET	/cases	Yes	All	Lists cases (scoped by role / assignment)	
	POST	/cases	Yes	Admin, Investigator	Creates new case dossier	
	GET	/cases/:id	Yes	All	Retrieves case details (enforces assignment)	
	PATCH	/cases/:id	Yes	Admin, Lead Inv	Modifies case metadata / status	
	GET	/cases/:id/comments	Yes	All	Lists threaded case comments	
	POST	/cases/:id/comments	Yes	Admin, Investigator	Posts comment with optional @mentions	
Evidence	POST	/cases/:caseId/evidence	Yes	Admin, Investigator	Streams evidence upload & calculates SHA-256	
	GET	/evidence/:id	Yes	All	Returns evidence metadata & logs ACCESSED	
	GET	/evidence/:id/download	Yes	All	Streams binary file download	
	POST	/evidence/:id/transfer	Yes	Admin, Custodian	Transfers custody to another user	
	GET	/evidence/:id/annotations	Yes	All	Returns coordinate annotations	
	POST	/evidence/:id/annotations	Yes	Admin, Investigator	Creates visual coordinate annotation	
Public	GET	/public/verify/:hash	No	Public	Zero-knowledge cryptographic hash lookup	
	POST	/public/verify-file	No	Public	Uploads file to verify SHA-256 match	
Reports	GET	/reports/evidence/:id/certificate	Yes	All	Generates court-admissible PDF	
certificate	GET	/reports/audit/csv	Yes	Admin, Auditor	Exports sanitized CSV audit logs	
Admin	GET	/admin/users	Yes	Administrator	Lists all user accounts	
	PATCH	/admin/users/:id/status	Yes	Administrator	Activates/deactivates user (mid-session cutoff)	
	PATCH	/admin/users/:id/role	Yes	Administrator	Modifies role (enforces last-admin logout)	

7. Evidence Lifecycle Walkthrough

```
[1. FIELD CAPTURE]
% Officer captures photo via mobile PWA camera (capture="environment")
% If offline: Queued in IndexedDB (evidence_drafts)
%

[2. STREAMING INTAKE & HASHING]
% Client streams multipart payload to POST /cases/:caseId/evidence
% Server streams binary to disk while piping through crypto.createHash('sha256')
% Calculated: 64-character lowercase hexadecimal digest
%

[3. STORAGE COMMIT & DB TRANSACTION]
% File saved under safe storageKey (e.g. /storage/ev-uuid.bin)
% Prisma transaction atomically creates:
% - Evidence record (status: PENDING, sha256: hash)
% - CustodyEvent (action: CREATED, actor: officer)
% - AuditLog record (action: EVIDENCE_UPLOADED)
%
```

```
[4. CUSTODY TRANSFER]ð
% Officer transfers evidence to Forensic Examiner:ð
% POST /evidence/:id/transfer { toUserId, note, reason }ð
% Verified: Caller is current custodianð
% Prisma updates currentCustodianId and logs CustodyEvent (action: TRANSFERRED)ð
%%ð

[5. ZERO-KNOWLEDGE PUBLIC VERIFICATION]ð
% Defense attorney or judge visits /verifyð
% Inputs SHA-256 hash or drops original fileð
% Backend queries Evidence table:ð
% - Found: Returns "VERIFIED", timestamp, and org name (redacts private dossier)ð
% - Altered: 1 byte changed -> Hash mismatch -> Returns "UNVERIFIED / TAMPERED"ð
%%ð

[6. COURT AUDIT EXPORT]ð
% Auditor requests official court documentation:ð
% GET /reports/evidence/:id/certificateð
% PDFKit renders tamper-proof PDF with chain of custody timeline and verification sealsð
```

8. Security Deep Dive ("How EviChain Stays Secure")ð

- 1. **Mid-Session Account Deactivation**: middleware queries the database on every authenticated request. Deactivated users are terminated immediately with 401 Unauthorized.ð
- 2. **Cross-Case IDOR Protection**: `getAuthorizedCase` middleware ensures only assigned lead investigators or Administrators can access or modify case dossiers.ð
- 3. **Server-Side Hashing Authority**: Client-reported hashes are completely ignored. The backend recalculates the SHA-256 digest from the uploaded file stream.ð
- 4. **Formula Injection Sanitization (CSV)**: All CSV export fields are sanitized by prefixing single-quotes to prevent malicious spreadsheet formula execution.ð
- 5. **Production Secrets Guard**: Process startup aborts if default development secrets are detected in production.ð

9. Mobile, PWA & Offline Field Capture Workflowð

- 1. **CJIS Cache Safety**: All private evidence endpoints are strictly marked `NetworkOnly` in `next.config.js` to prevent unencrypted evidence data from persisting in browser caches.ð
- 2. **IndexedDB Offline Storage**: When offline, files are saved in the client's IndexedDB database `evichain-offline` under object store `evidence_drafts`.ð
- 3. **Sync Engine**: When connection is restored, the client iterates drafts and streams uploads with unique idempotency keys, deleting drafts only after successful server confirmation.ð

10. Testing, Quality Gates & Regression Verificationð

```
13 automated test suites verifying all 13 modules with 194 passed tests:ð
• module2.test.ts (22 passed)ð
• module3.test.ts (17 passed)ð
• module4.test.ts (15 passed)ð
• module5.test.ts (17 passed)ð
• module6.test.ts (18 passed)ð
• module7.test.ts (16 passed)ð
• module8.test.ts (16 passed)ð
• module9.test.ts (12 passed)ð
• module10.test.ts (14 passed)ð
• module11.test.ts (12 passed)ð
• module12.test.ts (13 passed)ð
• e2e-recovery-smoke.ts (14 passed)ð
• module13.test.ts (8 passed)ð
Total: 194 / 194 (100% Pass Rate).ð
```

11. Local Setup & Production Operations Handbookð

- Step-by-step instructions for clean setup, database generation, supervisor startup, and health diagnostics.ð
- Terminal 1: `cd server && node supervisor.jsð`
 - Terminal 2: `npm run devð`
 - Health check: `http://localhost:4000/health/deepð`

12. System Debugging Playbook & Decision Trees

Covers port collisions (4000 in use), Neon PostgreSQL idle resets (10054), 401 mid-session rejections, 403 IDOR errors, and PWA camera fallbacks.

13. Safe Future Development Workflow

Branch hygiene, Prisma schema generation, rate limiting, and test regression standard maintenance.

14. Jury Demonstration Script & Hard Q&A Defense

6-minute demonstration walkthrough and direct answers to key questions:

- "Is EviChain a blockchain?" -> Cryptographic Immutable Ledger model.
- "What if an officer tampers locally?" -> SHA-256 hash mismatch flags file as TAMPERED immediately.
- "Can an Auditor change evidence?" -> Rejected with 403 Forbidden across 17 mutative routes.

15. Beginner Operator Manual (Hinglish & Simple Concepts)

Welcome! Agar aapne EviChain ko AI ki madad se build kiya hai aur aapko coding, terminal ya database ka deep experience nahi hai, toh yeh chapter aapke liye "A to Z Guide" hai.

1. EviChain Kya Hai? (Real-World Analogy)

EviChain police aur forensic teams ke liye ek "Digital Locker aur Register" hai:

- Evidence File = Sealed Parcel:** Crime scene se li gayi photo, CCTV footage, audio recording, ya PDF report.
- SHA-256 = Parcel ka Tamper-Proof Seal Number:** Ek 64-character ka unique digital fingerprint. Agar file ke andar 1 dot ya 1 space bhi change ho gaya, toh yeh seal number poora badal jayega!
- Chain of Custody = Signed Handover Register:** Ek register jisme record hota hai ki yeh parcel sabse pehle kisne collect kiya, kis tareekh ko kis forensic lab ko diya, kisne check kiya, aur ab kiske paas hai.
- Audit Log = CCTV Camera of System:** System me kisne kab login kiya, kya download kiya, aur kya update kiya, iska permanent record.

2. EviChain Me Kaun Kaun Se Users Hote Hain? (Roles)

Role Name	Simple Meaning	Real-World Role	Code Me Kya Kar Sakta Hai	Kya NAHI Kar Sakta
Administrator	System Boss / IT Officer	Station In-Charge / Admin	Naye users create karna, roles change karna, system settings dekhna, all cases view	Last admin account ko delete ya demote nahi kar sakta
Investigator	Field Police Officer / IO	Investigating Officer (IO)	Evidence upload karna, Custody transfer karna, Notes likhna	Admin settings nahi badal sakta, dusre officer ke private case me enter nahi ho sakta
Auditor	Independent Inspector / Court Examiner	Legal Inspector / Judge	Court report PDF aur CSV download karna, evidence verify karna	Saare audit logs dekhna, comment, ya transfer nahi kar sakta (403 Forbidden)
Public Visitor	Aam Nagrik / Defense Lawyer	Public Citizen	/verify page par SHA-256 hash paste karke check karna ki file original hai ya tampered	Case details, officer name ya private file download nahi kar sakta

3. EviChain Kya Guarantee Karta Hai aur Kya NAHI?

- Guarantee Karta Hai:** Data ki traceability, cryptographic immutability (koi file chupke se badal nahi sakti), aur signed chain of custody timeline.
- Legal Reality (Court Admissibility):** EviChain evidence ko verify karne ka tool hai; par kisi evidence ko court me accept karna ya na karna judge aur legal rules (jaise Section 65B Indian Evidence Act / BSA) par depend karta hai.

16. Daily Click-by-Click Workflows (All Screens & Menus)

Screen 1: Login & Registration (/login)

- Kahan milega:** Browser me `http://localhost:3000/login`
- Mode Toggle:** Screen par "Sign In" aur "Create Account" toggle hota hai.
- Registration:** Naya account create karne ke liye Name, Email, Password, aur Role select karein (Administrator, Investigator, Auditor, Custodian).
- Backend me kya hota hai:** Password bcrypt se hash hota hai, database me User create hota hai, aur 15-minute ka JWT access token return hota hai.
- Success:** Screen turant /dashboard par redirect ho jati hai.

Screen 2: Dashboard (/dashboard)

- Kahan milega:** Sidebar me "Dashboard" (Icon: 🏠).
- Kya dikhta hai:**
 - Metric Cards:** Total Active Cases, Registered Evidence Items, Pending Custody Transfers, Integrity Health.
 - Recent Activity Feed:** Haal hi me upload kiye gaye evidence aur transfer events.
 - Quick Action Buttons:** "New Case", "Upload Evidence", "Public Verify".

Screen 3: Create a Case (/cases/new)ð

- **Kaun kar sakta hai**: Administrator aur Investigator.ð
- **Kahan click karein**: Sidebar -> "Cases" -> Top right "+ New Case" button.ð
- **Form Fields**:ð
 - **Title**: Case ka naam (e.g. FIR-2026-CyberTheft-089).ð
 - **Description**: Case ke details aur initial findings.ð
 - **Status**: Active, Review, Closed, ya Archived (Default: Active).ð
 - **Priority**: Low, Medium, High, ya Critical (Default: Medium).ð
 - **Submit karne par**: Backend me Case record create hota hai jisme current user automatically **Lead Investigator** ban jata hai. Browser turant /cases/:id dossier par le jata hai.ð

Screen 4: Upload Evidence (/evidence/new)ð

- **Kahan click karein**: Sidebar -> "Evidence" -> "+ Upload Evidence".ð
- **Form Fields**:ð
 - **Select Case**: Kis case se evidence link karna hai.ð
 - **Evidence Name**: File ka descriptive label (e.g. CrimeScene_Camera1_Footage.mp4).ð
 - **Owner Organization**: Default Digital Forensics.ð
 - **Drag & Drop File**: Photo, Video, Audio, Document, ya Zip file select karein (Max 50MB).ð
 - **Backend Streaming Pipeline**:ð
 1. Multer file stream accept karta hai.ð
 2. Server crypto.createHash('sha256') se file ka streaming digest calculate karta hai.ð
 3. File safe storage key par save hoti hai (e.g. storage/uploads/ev-uuid.bin).ð
 4. Database me Evidence row, CustodyEvent (action: CREATED), aur AuditLog create hota hai.ð
 - **Success Result**: Screen par green badge dikhega aur 64-character SHA-256 fingerprint display hoga jise aap copy kar sakte hain.ð

Screen 5: Transfer Custody (/evidence/:id)ð

- **Kaun kar sakta hai**: Current Custodian ya Administrator.ð
- **Kahan click karein**: Evidence Detail page par "Transfer Custody" button.ð
- **Fields**:ð
 - **Recipient Officer**: Dropdown se naye custodian ko select karein.ð
 - **Transfer Reason / Note**: Handover ka reason (e.g. Transferred to Cyber Lab for forensic mobile extraction).ð
 - **Confirm**: "Confirm Transfer" par click karein.ð
 - **Result**: Evidence ka currentCustodianId update hota hai, CustodyEvent (TRANSFERRED) record hota hai, aur recipient ko real-time notification milta hai.ð

Screen 6: Public Zero-Knowledge Verification (/verify)ð

- **Kahan milega**: http://localhost:3000/verify (No login required!).ð
- **Kaise use karein**:ð
 - **Option A (Hash Mode)**: 64-character SHA-256 hash paste karein aur "Verify Hash" dabayein.ð
 - **Option B (File Mode)**: Koi bhi original file drag-and-drop karein.ð
 - **Result Output**:ð
 - **VERIFIED (Green)**: Status VERIFIED, intake timestamp, aur organization dikhayega. Private dossier ya officer ka naam public ko bilkul nahi dikhta!ð
 - **UNVERIFIED / TAMPERED (Red)**: Agar file me 1 byte bhi badla hai, toh system kahega "No matching evidence fingerprint found".ð

Screen 7: Court-Admissible Reports (/reports)ð

- **Kahan click karein**: Sidebar -> "Reports" (Icon: %¥).ð
- **Actions**:ð
 - **Download Certificate PDF**: Kisi bhi evidence item ka formal Forensic Integrity Certificate generate karein (PDFKit se A4 tamper-evident layout).ð
 - **Export Audit Ledger CSV**: Saare system events ko sanitized CSV me export karein.ð

17. Mobile / Offline Field Manual (PWA & IndexedDB Queue)ð

Crime scene par aksar network range nahi hoti. EviChain iske liye **Offline PWA Engine** use karta hai:ð

1. **Camera Open Karna**: Mobile browser me http://localhost:3000/mobile/evidence/camera kholein.ð
2. **Permissions**: Browser camera permission maangega ("Allow Camera"). Use grant karein.ð
3. **Rear Camera**: System automatically phone ke back/environment camera ko activate karta hai.ð
4. **Offline Mode**: Agar internet chala jaye, toh screen par yellow banner aayega: "Offline Mode — Evidence queued in local storage".ð
5. **Draft Save**: Photo capture karne par photo browser ke **IndexedDB** (evidence_drafts) me save ho jati hai.ð
6. **Sync Jab Internet Aaye**: Jab phone wapas Wi-Fi ya cellular me aata hai, Offline Queue Panel automatically active hota hai aur evidence ko server par stream karta hai. Jab server 201 Created bhej deta hai, tabhi local draft clear hota hai.ð

Draft Status Definitionsð

- **DRAFT**: File phone me temporarily save hai.ð
- **QUEUED**: File upload ke liye line me hai.ð
- **UPLOADING**: File server par stream ho rahi hai.ð
- **SYNCED**: Server ne SHA-256 calculate kar ke record register kar liya.ð
- **FAILED**: Network error ki wajah se upload fail hua; "Retry" button dabakar dobara koshish karein.ð

18. Operator Troubleshooting Decision Treeð

Problem (Aapko Kya Dikh Raha Hai)	Asli Wajah (Why It Happened)	Pehla Safe Step	Exact Command / Check
Kab Help Maangni Hai	Backend API server (Port 4000) band hai	Backend supervisor start karein	<code>cd server && node supervisor.js</code>
Unable to connect to EviChain API	Agar restart ke baad bhi fail ho	Agar restart ke baad bhi fail ho	
Port 4000 already in use	Purana node process abhi bhi background me chal raha hai	Stale process kill karein	<code>netstat -ano findstr :4000 -> taskkill /F /PID <PID></code>
Database error (10054 / connection reset)	Neon Cloud database ne idle connection drop kar diya	5 second wait karke refresh karein	<code>curl http://localhost:4000/health/deep</code>
Supervisor auto-reconnect karega	continuous error aaye toh internet check karein		
Login ke baad 401 Unauthorized	Account deactivate ho chuka hai ya JWT secret change hua hai	Admin panel -> Users -> Check Status	
Administrator se account active status check karayein	Agar Admin account bhi 401 de raha ho		
403 Forbidden on Case / Evidence	IDOR Protection: Aap us case ke assigned officer nahi hain	Case lead ya Administrator se access grant karayein	
Log in as Administrator to inspect	Agar aap assigned officer hain fir bhi 403 aaye		
PWA Camera Open Nahi Ho Raha	Desktop browser me webcam nahi hai ya HTTPS/localhost nahi hai	Chrome DevTools me device toolbar enable karein	
Chrome -> F12 -> Toggle Device Emulation	Hardware camera error par		

19. Demo-Day Checklist (Before, During & After)ð

Before Live Demo (1 Hour Pehle):ð

- Terminal 1: Backend supervisor live hai (`cd server && node supervisor.js`).
- Terminal 2: Next.js frontend live hai (`npm run dev`).
- Browser me check karein: `http://localhost:4000/health/deep` -> `{"ok": true, "database": "connected"}`.
- Do accounts ready rakhein:ð
 - Admin: `admin@evichain.gov` / `Admin@123` [VERIFY THIS IN YOUR LOCAL .env OR SEED DATA]ð
 - Investigator: `investigator@evichain.gov` / `Investigator@123` [VERIFY THIS IN YOUR LOCAL .env OR SEED DATA]ð
- Desktop par ek sample image/PDF ready rakhein upload test ke liye.ð

During Live Demo:ð

- Pehle problem explain karein (paper logs and manual tampering).ð
- File upload karein aur streaming SHA-256 fingerprint dikhayein.ð
- Incognito me `/verify` portal kholkar wahi hash verify karke dikhayein.ð
- 1 byte alter karke TAMPERED alert dikhayein.ð

20. Project Explanation Cheat Sheet (Speaking Guide)ð

30-Second Elevator Pitch:ð

“EviChain ek digital evidence management system hai jo crime scene se court room tak digital evidence ki integrity aur chain of custody ko guarantee karta hai. Yeh har file ka streaming SHA-256 fingerprint calculate karta hai, unalterable custody handover record karta hai, aur courts aur citizens ko bina private case data leak kiye zero-knowledge public verification ki suvidha deta hai.”ð

60-Second Explanation:ð

“High-profile criminal cases me aksar police par evidence tamper karne ya badalne ke aarop lagte hain. Traditional policing me paper registers aur unencrypted pen drives use hoti hain. EviChain is problem ko solve karta hai. Jaise hi koi officer photo ya CCTV footage upload karta hai, hamara backend streaming SHA-256 se uska digital fingerprint lock kar deta hai. Uske baad har view, har download, aur har officer-to-officer custody handover ek immutable audit ledger me record hota hai. Jury ya defense lawyer hamare public portal par sirf hash paste karke verify kar sakte hain ki evidence unhampered hai ya tampered.”ð

2-Minute Deep Pitch:ð

“Technical perspective se, EviChain ek decoupled Next.js 16 aur Node.js Express architecture par built hai, jise Neon PostgreSQL aur Prisma ORM power karte hain. Security hamari core foundation hai:ð

- Client-side se bheje gaye kisi bhi hash par trust nahi kiya jata; server khud byte-by-byte SHA-256 digest calculate karta hai.ð
- Cross-case IDOR defense aur mid-session deactivation enforcement unassigned officers ko doosre cases me jhaankne nahi deta.ð
- Field officers ke liye offline PWA camera workflow hai jo connectivity drop hone par evidence ko safely IndexedDB me queue kar leta hai.ð
- Poora system 194 passing automated tests aur auto-healing supervisor architecture ke saath enterprise-grade resilient hai.”ð

21. Appendixð

Architecture Glossaryð

- DEMS**: Digital Evidence Management System.ð
- SHA-256**: Secure Hash Algorithm generating a 256-bit (64-character hex) cryptographic fingerprint.ð
- IDOR**: Insecure Direct Object Reference (prevented via `getAuthorizedCase`).ð
- PWA**: Progressive Web App with offline caching and native device camera hooks.ð

- **Zero-Knowledge Verification**: Confirming the authenticity of an evidence file without exposing confidential case dossiers.

Verified vs Unimplemented Claims Matrix

- **Next.js 16 App Router**: Verified in Code.
- **Express 5 + TypeScript**: Verified in Code.
- **Neon PostgreSQL + Prisma ORM**: Verified in Code.
- **Server-Side SHA-256**: Verified in Code.
- **PWA & Offline IndexedDB**: Verified in Code.
- **Court PDF Export (PDFKit)**: Verified in Code.
- **Ethereum / Hyperledger Blockchain**: NOT IMPLEMENTED (Do not claim; state Cryptographic Immutable Ledger).
- **Hardware FIDO2 / WebAuthn**: NOT IMPLEMENTED (Auth uses bcrypt + JWT).

End of EviChain Complete Project Study Guide & Operator Manual

